

# All fixed order asymptotics for restricted two covers and line graphs

Research report 224

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Reconstructed edition 2

## Abstract

Restricted two-covers admit a positive Poisson sum whose unique real saddle gives an expansion to every fixed order. We prove a relative  $O_K(n^{-K-1})$  remainder at the exact saddle, then transfer the result through arbitrary fixed polynomial exponential multipliers, including the signed correction for labelled line graphs. A separate finite rational construction gives explicit Lambert- $W$  coefficients with the logarithmic remainder that their growth requires. We compute the first corrections, explain the distinction between the corrected line-graph sequence and the legacy formula recorded as A132219, and derive inverse expansions with a precise separation between model roots and certified integer thresholds. Exact arithmetic, independent endpoint enumeration, and reproducible TeX and code accompany the report. The previously known leading equivalents and exact identities are credited to Cameron, Prellberg and Stark; no claim of worldwide priority is made for the higher-order refinements.

[write] **Status and trust boundary, added 7 October 2026, batch 113.** This is bundle Report 224 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a060053-two-covers-line-graphs** (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.5). It is unrefereed and not formalized. The delivered text calls itself “Reconstructed edition 2” (5 October 2026); no earlier edition was delivered. Its author line reads “Research report 224” and its PDF author field “Research report”: it names no person or tool. *What rests on what.* The Poisson sum (1) and the identities (2) are proved in the text (and are Cameron, Prellberg and Stark’s); the corrected line-graph multiplier (4) rests on their classification of the exceptional roots (Whitney, Sabidussi), which the source imports and does not reprove. The asymptotic Theorems S1, S3 and C1 and the inverse statements of Section 4 have conventional proofs (strict concavity, Euler–Maclaurin and Poisson summation, Taylor and Cauchy estimates, the mean value theorem); no proof uses a computation, and the software and the floating diagnostics corroborate. The write adds Remarks 1 and 2 and dated notes; no statement, proof or number of the source was changed.

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# 1 Models and exact counting identities

## 1.1 Objects counted and notation

A two-cover of  $[n] = \{1, \dots, n\}$  is an unordered multiset of *nonempty* subsets, called blocks, in which each label occurs in exactly two block occurrences. It is restricted if distinct block occurrences intersect in at most one label. It is proper if no block is repeated. The only repetitions possible in a restricted cover are repeated singleton blocks. Write  $U_n$  for all restricted covers and  $V_n$  for proper restricted covers. The empty cover has index zero and count one. The sequences are respectively A014500 and A060053 [4, 3].

Turn each block occurrence into a vertex and each label into an edge between the two blocks containing it. This gives a simple graph with labelled edges, unlabelled vertices and no isolated vertices. Conversely the incident edge-label sets at its vertices recover the cover. A repeated singleton is exactly an isolated-edge component. Thus  $V_n$  forbids isolated edges and  $U_n$  permits them. A *root graph* of a line graph means a graph whose line graph is the specified graph; no vertex is distinguished by this terminology.

Let  $L_n$  count simple line graphs with vertex set  $[n]$ . Their root graphs have labelled edges and unlabelled nonisolated vertices. Different roots, and different edge labellings on the same unlabelled root, can give the same labelled line graph. This matters in the small connected exceptions discussed below.

All capital-letter generating series in this report are formal exponential generating series. In particular,  $H(x) = \sum H_n x^n / n!$  is not an analytic function at positive  $x$ . The individual sums defining its coefficients do converge.

## 1.2 The positive Poisson sum

Put  $M(t) = t(t-1)/2$  and use  $(z)_n = z(z-1)\cdots(z-n+1)$  for a falling factorial, with  $(z)_0 = 1$ . Define

$$H_n = e^{-1} \sum_{m \geq 0} \frac{(M(m))_n}{m!}. \quad (1)$$

At integer  $m$ , any term with  $M(m) < n$  is zero. There is no sign cancellation in this sum. Its convergence follows from the factorial denominator and the polynomial numerator. For  $n = 0$  it equals one.

If a root graph  $G$  has  $v$  vertices and  $k$  isolated-edge components, its edge-label-preserving automorphism group has order  $2^k$ . Indeed, a vertex incident with at least two distinct labelled edges is fixed because it is their unique common endpoint. Adjacent leaves are then fixed; the only remaining freedom is to interchange the endpoints of each isolated edge. Among graphs on  $m$  labelled vertices, the number of embeddings of  $G$  is  $(m)_v / 2^k$ . As  $(M(m))_n$  counts ordered choices of  $n$  distinct edges of  $K_m$ , summing over  $m$  gives

$$e^{-1} \sum_{m \geq v} \frac{(m)_v}{2^k m!} = 2^{-k}.$$

Consequently  $H_n$  is the sum of the weights  $2^{-k}$  over all the  $U_n$  roots. Decomposing into isolated edges and the remaining proper cover proves

$$H(x) = e^{x/2} V(x), \quad U(x) = e^x V(x) = e^{x/2} H(x). \quad (2)$$

These are also the exact identities in [1, Proposition 3 and equation (18)]. This argument explains why  $H_n$  is generally rational, whereas  $U_n$  and  $V_n$  are integers.

There is a useful exact finite alternative to the infinite sum. If

$$\prod_{j=0}^{n-1} (t^2 - t - 2j) = \sum_{k=0}^{2n} c_{n,k} t^k,$$

then the Poisson moment identity for the Bell numbers  $B_k$  gives

$$H_n = 2^{-n} \sum_{k=0}^{2n} c_{n,k} B_k. \quad (3)$$

Both the polynomial multiplication and Bell numbers can be evaluated using integers. Equation (3) is an exact algorithm, not an asymptotic replacement.

### 1.3 The corrected line graph multiplier

The corrected Proposition 5 of Cameron, Prellberg and Stark [1, p. 6], using the Whitney–Sabidussi line-graph results, gives

$$\begin{aligned} L(x) &= \exp \left( -\frac{x^3}{6} - \frac{x^4}{4} - \frac{x^5}{8} - \frac{x^6}{48} \right) U(x) \\ &= \exp \left( \frac{x}{2} - \frac{x^3}{6} - \frac{x^4}{4} - \frac{x^5}{8} - \frac{x^6}{48} \right) H(x). \end{aligned} \quad (4)$$

We use their classification of the exceptions as a known theorem. Here is the labelled-count calculation that explains the four coefficients. At three edges the triangle and the three-edge star give two roots with one common labelled line image, a difference of one. The remaining exceptional connected roots are the paw ( $K_4$  minus two adjacent edges),  $K_4$  minus one edge, and  $K_4$ . Their root and line-image counts are respectively 12 and 6, 30 and 15, and 30 and 15. The differences 1, 6, 15, 15, divided by  $3!, 4!, 5!, 6!$ , are the four exponents in (4). The exponential formula passes from the connected differences to all graphs. Completeness of this list is the cited structural theorem, not an inference from small computations.

The original arXiv version [2, p. 6] omitted the last three terms. Its formula

$$O(x) = e^{-x^3/6} U(x)$$

will be called the *legacy* series. The A132219 snapshot retrieved on 5 October 2026 matches this legacy formula in all 18 available terms, indexed 1 through 18, rather than the corrected count. The corrected author manuscript explicitly records the correction and acknowledges the referee’s observation. We have not independently retrieved the typeset journal PDF, so source-page locators here refer to the author manuscripts.

| $n$ | $2^n H_n$ | $V_n$ | $U_n$ | $L_n$ | $O_n$ |
|-----|-----------|-------|-------|-------|-------|
| 0   | 1         | 1     | 1     | 1     | 1     |
| 1   | 1         | 0     | 1     | 1     | 1     |
| 2   | 5         | 1     | 2     | 2     | 2     |
| 3   | 53        | 5     | 9     | 8     | 8     |
| 4   | 873       | 43    | 70    | 60    | 66    |
| 5   | 20457     | 518   | 794   | 729   | 774   |
| 6   | 634541    | 8186  | 12055 | 11600 | 11885 |

Already  $O_4 = 66$  exceeds the  $2^{\binom{4}{2}} = 64$  labelled simple graphs on four vertices. Thus it cannot count line graphs as defined here. Independent enumeration of partitions of the  $2n$  labelled edge endpoints verifies the weighted endpoint count  $2^n H_n$  and the counts  $U_n, V_n, L_n$  through  $n = 6$ .

[write] **Seven edges, 7 October 2026.** The write's enumeration of restricted two-covers from the definition gives the table's  $U_n, V_n$  and  $L_n$  again and extends it by  $U_7 = 233,238$ ,  $V_7 = 163,356$  and  $L_7 = 228,443$  (the last as the number of distinct labelled line images, without the exception classification), all equal to (4) and the identities (2). The column  $2^n H_n$  is OEIS A094089 (Remark 2), whose e.g.f.  $e^{-1} \sum_m (1+2x)^{\binom{m}{2}} / m!$  is  $H(2x)$ . On four vertices the only labelled graphs that are not line graphs are the four claws, so  $L_4 = 64 - 4 = 60$ .

## 1.4 Prior results and the scope of the refinement

Write  $r = W(2n)$  for the positive Lambert function, so  $re^r = 2n$ . The leading equivalent

$$V_n \sim U_n \sim L_n \sim H_n \sim 2^{-n} B_{2n} \exp(-r^2/4 - r/2) \quad (5)$$

is already contained in the analysis of [1, pp. 17–19, equations (21)–(27)]. Their Theorem 1 states an equivalent logarithmic form. The leading equivalent, Poisson identity and exact graph multipliers are prior results. Our purpose is to give a self-contained fixed-order saddle analysis, computable explicit refinements and inverse error control. Historical coverage is not exhaustive enough to support a worldwide novelty claim.

The main distinctions are substantive. An exact-saddle remainder can be  $O(n^{-K-1})$  because the local derivatives already incorporate the logarithmic scale. An explicit Lambert expansion instead has coefficients growing in  $r$ , and its displayed remainder retains that growth. A signed polynomial convolution has a third error scale. We prove each estimate in its own normalization rather than transfer a remainder by notation alone.

## 1.5 Provenance, sources and reading conventions

[write] **Added 7 October 2026, batch 113.** *Provenance.* This report is Report 224 of the session bundle that ProveIt's docs/incoming intake processed as batch 113: the archive `Report224.zip` (621,206 bytes, 27 files, wrapper directory `Report224/`), arrival commit `60f54ea06`, placed in `a4186a946`. The text is the delivered `src/report224.tex` (44 lines) with its six `\input` files `models.tex`, `saddle.tex`, `coefficients.tex`, `inverses.tex`, `computation.tex` and `references.tex` (1,383 lines in all, 21 pages, dated 5 October 2026), shipped as `article.tex` and `sections/*.tex`; the write changed the six `\input` paths accordingly. The programs are shipped under `code/` (the delivered root `build.py` as `code/build.py`); the data, the build receipt `build_checks.json` and the three retrieved b-files (as `data/sources-A*_bfile.txt`) under `data/`; the delivered `SOURCES/SOURCE_LEDGER.md` as `SOURCE_LEDGER.md`. Not shipped, and retrievable from the arrival commit: the delivered `Report224.pdf`, the checksum manifest `MANIFEST.sha256` (24 entries, all verified at the write) and the delivered `README.md`, which the report README replaces. The package records no ProveIt commit; it carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 47 delivered labels with `tc1:` and updated the references to them (`3 \eqref`); it added this subsection, the notes marked [write] and Remarks 1 and 2, each the last statement of its section, and inserted no numbered display, so every section, equation and tag of the source ((1)–(5), (S1)–(S27) with (S21a), (C1)–(C23)) keeps its delivered number. The delivered text uses no theorem environment: its Theorems S1, S3, C1 and Proposition S2 are named in bold type, so the write's remarks are numbered 1 and 2. *The sources, as the write read them* (7 October 2026): OEIS A060053, A014500, A132219 and A094089 in the internal format, with their b-files (Remark 2); Cameron, Prellberg and Stark's corrected author manuscript `cover.pdf` and their arXiv version 1, retrieved from the ledger's URLs (196,940 and 175,905 bytes; both SHA-256 values equal those in `SOURCE_LEDGER.md`): Proposition 3 (p. 5), the corrected Proposition 5 (p. 6, the multiplier  $\exp(-x^3/3! - 6x^4/4! - 15x^5/5! - 15x^6/6!)$ , which is (4), with its proof from Whitney's

theorem and Sabidussi's result), equation (18) (p. 16), equations (21)–(27) (pp. 17–18), the formula for  $L(x)$  and the acknowledgement of the referee's correction (p. 19), and Proposition 5 of version 1 (p. 6,  $L(x) = e^{-x^3/3!}U(x)$ ): every locator of the source and of its ledger is as stated. Also the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (p0:def:model, p0:thm:lambert-core, p0:def:three-inverses, p0:thm:staircase, plt:def:lw-monomial-log-datum, plt:thm:lw-template). Not read by the write: the typeset journal version (the source did not obtain it either), the DLMF sections (standard conventions), and the OEIS revision histories that the ledger reports having inspected.

*What was checked at intake.* The batch-113 dossier read the manuscript, found no demonstrably wrong claim, recorded that A132219 cannot count line graphs ( $a(4) = 66 > 2^6$ ), and reran the delivered suite on a copy (it reproduces). *What the write checked.* It read every proof and found no error. With its own code (Python 3.14.4, exact integers and fractions; SymPy 1.14.0; mpmath 1.3.0 at 50–60 digits), not importing the delivered programs, it

- computed  $H_n$  by two exact routes, the Newton forward differences of  $(M(m))_n$  (no Bell numbers;  $n \leq 120$ ) and the Bell-moment formula (3) ( $n \leq 300$ ), which agree;  $2^n H_n$  is an integer for  $n \leq 300$ ;  $V_n$ ,  $U_n$ ,  $L_n$ ,  $O_n$  from the multipliers are integers for  $n \leq 300$ , equal to the table of Section 1.3, to all 101 terms of the b-files of A060053 and A014500, to the 18 terms of A132219 (legacy) and, for  $2^n H_n$ , to all 151 terms of the b-file of A094089;  $U$  is the binomial transform of  $V$  for  $n \leq 300$ ;
- enumerated, directly from the definition, all restricted two-covers of  $[n]$  for  $n \leq 7$  (multisets of blocks, each label in two occurrences, occurrences meeting in at most one label) and their distinct labelled line images:  $U_n$ ,  $V_n$  and  $L_n$  equal the formulas through  $n = 7$ , in particular  $L_7 = 228,443$  (one edge beyond the source's endpoint check, and without the exception classification); the root and line-image counts 1, 1; 12, 6; 30, 15; 30, 15 of the triangle and star, paw,  $K_4$  minus an edge and  $K_4$  by automorphism counts;
- rederived the Lambert coefficients  $a_{P,j}$ ,  $j \leq 3$ , for  $H, V, U, L$  and the Bell normalization with its own SymPy expansion of the saddle sum ( $m = s + \sqrt{s}y$ ,  $n = r/(2h^2)$ , finite Faulhaber, Stirling and collision truncations): equal to every coefficient printed in Section 3.4 and to all of `data/coefficients_order3.json`;  $E_1$ ,  $E_2$  and the first-order formula of Section 5.2;  $v_1$ ,  $v_2$ ,  $v_3$  and  $v_3 \sim r^{12}/663552$ ; (C16) ( $a_{L,3} - a_{U,3} = -r^6/48$ ); the contraction coefficients (S3) and the vanishing of the odd orders; (C17); the cancellation defining  $d_1$  in (C20); the exact form  $e^{r_0}(r_0^2 - ar_0 + 1)$  of  $\Phi(x_0)$  and the expansion (C23) (through  $w^{-6}$ ; the next term is  $b^2/(2w^4)$ , so its  $O(w^{-4})$  is sharp);
- evaluated  $H_m$  for  $n - 8 \leq m \leq n$ ,  $n = 10^4, \dots, 10^7$ , from the positive sum (1) and  $B_{2n}$  from Dobinski's sum: for each of  $H, V, U, L$  and each  $j \leq 3$ ,  $n^{j+1}$  times the remainder after the  $j$ -th Lambert term approaches  $a_{P,j+1}(r)$  (e.g. for  $H$  at  $n = 10^7$ :  $634.597$ ,  $174718.3$ ,  $2.72990 \times 10^7$  against  $634.579$ ,  $174715.6$ ,  $2.72987 \times 10^7$ ), and the remainder after  $j = 3$  is at most  $2 \times 10^{-9} r^{16}/n^4$  in absolute value;  $V_n/T_n$  against  $v_1, v_2, v_3$  (ratios 1.0000246, 1.0000132, 1.0000074 at  $n = 10^7$ );  $48n^3(1 - L_n/U_n)/r^6 = 1.0063$ , 1.00093, 1.00013, 1.000017; and  $(H_{n-1}/H_n)/(r^2/(2n^2)) \rightarrow 1$  (S21);
- recomputed the exact-saddle table of Section 5.2 at  $n = 1000$  ( $\tau$ ,  $\sigma$ ,  $\lambda_3, \dots, \lambda_8$  from (S7),  $C_0, \dots, C_3$ ): every printed value (note in Section 5.2).

It reran, on a fresh copy of the delivered layout, `test_exact.py` with `--cap 640` (normal and -0), `endpoint_check.py` with `--max-index 6`, `verify_coefficients.py`, `generate_coefficients.py` at orders 2 and 3 and `check_saddle.py` with the frozen parameters: both test runs pass and print exactly the exact block of `data/build_checks.json` (cap 640, the 101, 101 and 18 retrieved terms, the count-data hash); the endpoint run equals `data/endpoints6.json`; the verifier passes its 28 identity checks; the generated coefficient files equal `data/coefficients_order2.json` and `data/coefficients_order3.json`; and `check_saddle.py` reproduces `data/saddle_diagnostics.json` exactly. The PDF and ZIP builder was not run. Floating computations are observations, not certificates.

*Relation to the repository.* Before batch 113 no file of the repository named A060053, A014500, A132219 or A094089. The report `a307316-leafless-multigraphs` cites Cameron–Prellberg–Stark for the labelled-edge multigraph reading of two-covers, for different objects (unlabelled leafless loopless multigraphs); no result is shared and no reciprocal note is proposed. No Lean or Rocq file treats these sequences; no statement of this

report is formalized, and its place in the collection confers no formal status. The inverse is compared with the transseries volume in Remark 1.

**[write] What is not claimed, collected from the source.** The exact identities, the Poisson sum, the exception classification for line graphs and the leading equivalent (5) are prior (Cameron–Prellberg–Stark); the classification is used as a known theorem, and the small enumerations validate its consequences, not its completeness. No worldwide novelty or priority claim for the higher-order refinements; source coverage is not an exhaustive historical search; no conjecture or remainder is attributed to OEIS. The software caps (exact  $n \leq 640$ , symbolic order  $\leq 3$ , numerics  $n \leq 10^4$ ) are implementation limits, not limits of the theorems; the floating diagnostics are not interval certificates and give no effective constant or starting index; the  $O$ -constants are not effective; a model root does not certify an arbitrary integer threshold without a true-sequence enclosure; the exact threshold scan is not an unbounded inverse oracle; the Lambert remainder carries powers of  $r$  and is not  $O(n^{-3})$  with a constant independent of  $n$ ; byte reproducibility is claimed only for the same toolchain. *The write adds:* its checks are exact or floating computations; its transseries remark claims no novelty for any inversion.

*Reading conventions* (letters the source uses in more than one sense, with the tempting false readings; no symbol was renamed).

| Symbol                    | Meanings                                                                                                                                                                                                                                                      |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a, a_j, a_{P,j}$         | $a_j = (1 + \sqrt{1 + 8j})/2$ the roots in Section 2.2; $a_j = \lambda_j/\eta^{j-2}$ in Section 2.4; $a_{P,j}(r)$ the Lambert coefficients (C2); $a = 1 + \frac{1}{2} \log 2$ in Section 4.2. <i>False reading:</i> $a_{P,j}$ is not a root $a_j$ .           |
| $b, b_j$                  | $b_j = 1 - a_j$ ; $b_j(r)$ the Bell corrections and the recurrence coefficients of $\exp(E)$ in Section 3.2; $b = 1 - a^2/4$ in (C23).                                                                                                                        |
| $B_k, \mathbf{B}_k$       | Bell numbers; sans-serif: Bernoulli numbers and polynomials (the source flags this).                                                                                                                                                                          |
| $c_k, p_j, c_{n,k}$       | $p_j = [x^j]e^{P(x)}$ (Section 2.6) and $c_k = [z^k]e^{P(z)}$ (Section 3.1) are the same numbers; $c_{n,k}$ the coefficients in (3); $c, C, C_q$ constants. <i>False reading:</i> $C_\ell$ in (S1) are the contraction coefficients, not constants.           |
| $E, \mathbb{E}$           | $\mathbb{E}$ expectation; $E$ the formal exponent (C4), $E_J$ its finite analytic version (Section 3.3.2) and, in Section 4.3, an error bound $E_J(k)$ ; $E_k$ the coefficients of $E$ . <i>False reading:</i> the bound $E_J(k)$ is not the exponent $E_J$ . |
| $F_p, F_J, f_n$           | $F_p(n)$ Faulhaber sums in (C3); $F_J$ the inverse model (C21); $f_n$ the saddle exponent.                                                                                                                                                                    |
| $H, h, h_0$               | $H_n, H(x)$ the weighted Poisson sum; $h = s^{-1/2}$ (Section 3.2); $h(u, x)$ the cubic-and-higher part in (S16); $h_0(r)$ in Section 4.1.                                                                                                                    |
| $L, \mathcal{H}$          | $L_n, L(x)$ labelled line graphs; $L = 2K + 4$ the Euler–Maclaurin order in (S14)–(S15); $\mathcal{H}_{m,K}$ the truncated exact-saddle expression.                                                                                                           |
| $M, m$                    | $M(t) = t(t-1)/2$ ; $M$ also the formal quantity of Section 3.2; $m$ the summation index and $m = x/h^2$ .                                                                                                                                                    |
| $O, O_n$                  | Landau symbol; $O(x), O_n$ the legacy series of Section 1.3 (A132219). <i>False reading:</i> $O_4 = 66$ is a count, not an error term.                                                                                                                        |
| $Q, q$                    | $Q = 2K + 1$ in Section 2.4; $Q, Q_J$ the multiplier (C5); $q$ a derivative order, the Stirling index in (C4), and $q(r) = 2r - \log 2$ in (C17).                                                                                                             |
| $r, s, S_n$               | $r = W(2n)$ , $s = e^r = 2n/r$ (also $r = W(2x)$ in Section 4); $S_n$ is defined twice, after (S21) and in (C1), with equal values. <i>False reading:</i> $r_0 = W(2x_0)$ , not $W(2N)$ .                                                                     |
| $T_n, x$                  | $T_n$ the exact Bell normalization; $x$ the e.g.f. variable, $x = (t - \tau)/\sigma$ in Section 2, $x = 1 + hy$ in Section 3 and the continuous index in Section 4.                                                                                           |
| $Y, y, w, \delta$         | $Y$ the centred Gaussian in (C6) and the target value in Section 4; $y$ the scaled saddle coordinate; $w, \delta$ the Lambert seed of (C23).                                                                                                                  |
| $P, \mathbb{P}_k, K, K_P$ | $P$ the polynomial perturbation; $\mathbb{P}_k$ the law in (S20); $K$ an order; $K_P$ the coefficient field of $P$ .                                                                                                                                          |

## 2 Exact saddle expansion and polynomial perturbations

### 2.1 Definitions and statement

Put

$$M(t) = \frac{t(t-1)}{2}, \quad H_n = e^{-1} \sum_{m \geq 0} \frac{(M(m))_n}{m!}, \quad H_0 = 1,$$

where  $(u)_n = u(u-1) \cdots (u-n+1)$ . For each fixed  $n$ , this numerical sum converges absolutely, since its numerator is a polynomial of degree  $2n$  in  $m$ . Because  $M(m)$  is a nonnegative integer, terms with  $M(m) < n$  vanish. For  $n \geq 1$ , define

$$a_j = \frac{1 + \sqrt{1 + 8j}}{2}, \quad b_j = 1 - a_j, \quad f_n(t) = \sum_{j=0}^{n-1} \log(M(t) - j) - \log \Gamma(t+1) \quad (t > a_{n-1}).$$

There is a unique maximum  $\tau = \tau_n$  of  $f_n$ . Define

$$r = W(2n), \quad s = \frac{2n}{r} = e^r, \quad \sigma^2 = -\frac{1}{f_n''(\tau)}, \quad \lambda_j = f_n^{(j)}(\tau) \sigma^j \quad (j \geq 3).$$

For  $\ell \geq 0$ , let

$$C_\ell = \sum_{\substack{k_3, \dots, k_{2\ell+2} \geq 0 \\ \sum_{j=3}^{2\ell+2} (j-2)k_j = 2\ell}} \left( \sum_j j k_j - 1 \right)!! \prod_{j=3}^{2\ell+2} \frac{(\lambda_j/j!)^{k_j}}{k_j!},$$

with the empty summand giving  $C_0 = 1$  and  $(-1)!! = 1$ .

**Theorem S1.** For every fixed integer  $K \geq 0$ ,

$$H_n = e^{f_n(\tau)-1} \sqrt{2\pi} \sigma \left( \sum_{\ell=0}^K C_\ell + O_K(n^{-K-1}) \right). \quad (\text{S1})$$

Moreover,

$$\tau = s + O(r), \quad \sigma^2 = \frac{s}{r+1} \left( 1 + O(r^2/n) \right), \quad \lambda_j = O_j(n^{1-j/2}), \quad C_\ell = O_\ell(n^{-\ell}). \quad (\text{S2})$$

In particular the error in (S1) has no unreported positive power of  $r$ .

The first two corrections are

$$C_1 = \frac{\lambda_4}{8} + \frac{5\lambda_3^2}{24},$$

$$C_2 = \frac{\lambda_6}{48} + \frac{7\lambda_3\lambda_5}{48} + \frac{35\lambda_4^2}{384} + \frac{35\lambda_3^2\lambda_4}{64} + \frac{385\lambda_3^4}{1152}. \quad (\text{S3})$$

These coefficients use derivatives at the exact saddle; replacing the saddle by  $s$  without expanding all accompanying factors changes the coefficients.

## 2.2 Strict concavity, localization, and derivative scale

The factorization

$$M(t) - j = \frac{1}{2}(t - a_j)(t - b_j)$$

gives

$$\begin{aligned} f'_n(t) &= \sum_{j < n} \left( \frac{1}{t - a_j} + \frac{1}{t - b_j} \right) - \psi(t + 1), \\ f''_n(t) &= - \sum_{j < n} \left( \frac{1}{(t - a_j)^2} + \frac{1}{(t - b_j)^2} \right) - \psi'(t + 1) < 0. \end{aligned} \quad (\text{S4})$$

The trigamma function is positive on this interval. Also  $f'_n(t) \rightarrow +\infty$  at the lower endpoint and  $f'_n(t) \rightarrow -\infty$  at infinity. This proves existence and uniqueness of  $\tau$ , with no assumption about the local approximation.

Uniformly for  $t \asymp s$ , Taylor expansion of the rational factors and the usual positive-real digamma estimates yield

$$f'_n(t) = \frac{2n}{t} - \log t + O\left(\frac{n}{t^2} + \frac{n^2}{t^3} + \frac{1}{t}\right), \quad (\text{S5})$$

$$f''_n(t) = -\frac{2n}{t^2} - \frac{1}{t} + O\left(\frac{n}{t^3} + \frac{n^2}{t^4} + \frac{1}{t^2}\right). \quad (\text{S6})$$

Here  $\max_j |a_j| + |b_j| = O(\sqrt{n}) = o(s)$ . At  $s/2$  and  $2s$ , the leading part of (S5) has opposite signs and size comparable to  $r$ , while the error is  $o(r)$ . Hence  $\tau \in (s/2, 2s)$ . At  $s$ , the leading part vanishes and

$$f'_n(s) = O(r^3/n), \quad |f''_n(t)| \asymp n/s^2 \asymp r^2/n$$

on the interval joining  $s$  to  $\tau$ . The mean value theorem gives  $\tau - s = O(r)$ . Substituting this into (S6) gives the variance estimate in (S2).

More generally, for each fixed  $q \geq 2$ ,

$$f_n^{(q)}(t) = (-1)^{q-1}(q-1)! \sum_{j < n} ((t - a_j)^{-q} + (t - b_j)^{-q}) - \psi^{(q-1)}(t + 1). \quad (\text{S7})$$

The polygamma series bounds its last term by  $O_q(t^{1-q})$ . Thus, uniformly on  $I_n = [\tau/2, 3\tau/2]$ ,

$$|f_n^{(q)}(t)| \leq C_q n/s^q, \quad -f''_n(t) \asymp n/s^2 \asymp \sigma^{-2}. \quad (\text{S8})$$

Since  $s(r+1) \asymp n$ , multiplying (S8) by  $\sigma^q$  gives

$$\sup_{t \in I_n} |f_n^{(q)}(t)| \sigma^q \leq C_q n^{1-q/2}. \quad (\text{S9})$$

This is the uniform derivative estimate responsible for the absence of logarithmic leakage. It also proves  $C_\ell = O(n^{-\ell})$ : each monomial has order  $n^{-\frac{1}{2} \sum (j-2)k_j} = n^{-\ell}$ .

### 2.3 Global tails and a smooth lattice cutoff

Integrating (S8) twice from the saddle gives, for  $t \in I_n$ ,

$$f_n(t) - f_n(\tau) \leq -c(t - \tau)^2/\sigma^2. \quad (\text{S10})$$

In particular, at  $3\tau/4$  and  $5\tau/4$ , the loss is at least  $cn$ , because  $\tau^2/\sigma^2 \asymp n$ . To the left, concavity and monotonicity bound the entire remaining interval by the left boundary value. To the right of  $3\tau/2$ , concavity gives the tangent bound

$$f_n(t) \leq f_n(3\tau/2) - c(n/\tau)(t - 3\tau/2), \quad t \geq 3\tau/2. \quad (\text{S11})$$

Here (S8), integrated once, shows  $f'_n(3\tau/2) \leq -cn/\tau$ . Equations (S10) and (S11) show that both the integral and the integer sum outside  $[3\tau/4, 5\tau/4]$  are

$$O\left((1 + \tau)e^{f_n(\tau) - cn}\right). \quad (\text{S12})$$

For the sum in the far right tail, sum the exponential tangent majorant as a geometric series; the far left contains  $O(\tau)$  integer points. This supplies a global argument, rather than extending a local quadratic estimate beyond its domain.

Choose a fixed smooth cutoff  $\chi$ , equal to one on  $[3/4, 5/4]$ , supported in  $(1/2, 3/2)$ , with  $0 \leq \chi \leq 1$ , and put

$$g_n(t) = \chi(t/\tau)e^{f_n(t)},$$

extended by zero to  $\mathbb{R}$ . For every fixed integer  $L \geq 0$ ,

$$\|g_n^{(L)}\|_1 \leq C_L e^{f_n(\tau)} \sigma^{1-L}. \quad (\text{S13})$$

To see this, write  $x = (t - \tau)/\sigma$ . On the support, (S8) and  $f'_n(\tau) = 0$  imply

$$|f'_n(t)| \leq C\sigma^{-1}|x|, \quad |f_n^{(q)}(t)| \leq C_q\sigma^{-q} \quad (q \geq 2).$$

The derivative formula for  $e^{f_n}$  therefore bounds its  $L$ -th derivative by  $C_L\sigma^{-L}(1 + |x|)^Le^{f_n(t)}$ . The cutoff derivatives cost only  $O(\tau^{-q}) = O(\sigma^{-q})$ . Integrating the resulting polynomial times the Gaussian majorant (S10) proves (S13).

For even  $L \geq 2$ , the compact-support Euler–Maclaurin remainder gives

$$\left| \sum_{m \in \mathbb{Z}} g_n(m) - \int_{\mathbb{R}} g_n(t) dt \right| \leq \frac{2\zeta(L)}{(2\pi)^L} \|g_n^{(L)}\|_1. \quad (\text{S14})$$

Dividing by  $e^{f_n(\tau)}\sigma$  leaves  $O_L(\sigma^{-L})$ . For the expansion through  $n^{-K}$ , take

$$L = 2K + 4.$$

Then

$$\sigma^{-L} = O\left(r^{2K+4}n^{-K-2}\right) = O_K(n^{-K-1}), \quad (\text{S15})$$

since every fixed power of  $\log n$  is  $O(n)$ . Taking only  $L = 2K + 2$  would leave  $r^{2K+2}n^{-K-1}$ , which does **not** justify the claimed remainder. Together with (S12), (S14)–(S15) rigorously replace the lattice sum by the continuous integral at the required accuracy.

## 2.4 Integrated Taylor expansion to every fixed order

Set  $\eta = n^{-1/2}$  and  $Q = 2K + 1$ . Taylor's theorem and (S9), through derivative order  $2K + 4$ , give

$$f_n(\tau + \sigma x) - f_n(\tau) = -x^2/2 + h(\eta, x) + R_n(x),$$

$$h(u, x) = \sum_{j=3}^{2K+3} a_j u^{j-2} x^j / j!, \quad a_j = \lambda_j / \eta^{j-2} = O_j(1),$$

$$|R_n(x)| \leq C_K \eta^{2K+2} |x|^{2K+4}. \quad (\text{S16})$$

These estimates hold throughout  $|x| \leq \delta/\eta$ , with a sufficiently small fixed  $\delta > 0$  chosen so the associated  $t$ -interval lies inside the region where the cutoff is one. Reducing  $\delta$  if necessary, uniformly for  $0 \leq u \leq \eta$ ,

$$|h(u, x)| \leq x^2/8, \quad |R_n(x)| \leq x^2/8. \quad (\text{S17})$$

Indeed each term of  $h$  is bounded by a constant times  $x^2(\eta|x|)^{j-2}$ , and the remainder by a constant times  $x^2(\eta|x|)^{2K+2}$ .

For fixed  $x$ , expand  $e^{h(u,x)}$  at  $u = 0$  through degree  $Q$ . The coefficients are precisely the monomials indexed by  $\sum (j-2)k_j \leq Q$ . The derivative of order  $Q + 1$  with respect to  $u$  is, by repeated differentiation,  $e^{h(u,x)}$  times a polynomial in derivatives of  $h$ . Because all  $a_j$  are bounded and  $0 \leq u \leq \eta \leq 1$ , this is bounded by

$$C_K(1 + |x|)^{D_K} e^{x^2/8}$$

for a finite fixed  $D_K$ . The Taylor remainder is therefore bounded by that expression times  $\eta^{Q+1}$ . Also, (S16)–(S17) and the mean value bound for the exponential imply

$$|e^{h(\eta,x)+R_n(x)} - e^{h(\eta,x)}| \leq C_K \eta^{2K+2} |x|^{2K+4} e^{x^2/4}.$$

Multiplication by  $e^{-x^2/2}$  gives an integrable polynomial times a fixed Gaussian. Hence the **integrated** total remainder is  $O_K(\eta^{2K+2}) = O_K(n^{-K-1})$ , without enlarging the error by the length of the central interval.

Outside  $|x| \leq \delta/\eta$ , (S10) bounds the cutoff integral by  $e^{-c_K n}$  times its leading scale. We may likewise extend each of the finitely many polynomial-Gaussian integrals to all of  $\mathbb{R}$ , with an exponentially small error. A monomial indexed by  $k_j$  has ordinary degree  $D = \sum j k_j$  and weighted degree  $w = \sum (j-2)k_j$ . Their parities agree. All odd weighted degrees therefore integrate to zero. For  $w = 2\ell$ , the normalized Gaussian moment is  $(D-1)!!$ , giving precisely  $C_\ell$ .

Thus

$$\int g_n(t) dt = e^{f_n(\tau)} \sqrt{2\pi\sigma} \left( \sum_{\ell=0}^K C_\ell + O_K(n^{-K-1}) \right).$$

Combining this with (S12)–(S15) and the defining factor  $e^{-1}$  proves Theorem S1.

## 2.5 Growth and an adjacent-ratio bound from positivity

**Proposition S2.**

$$\log H_m = 2m \log m - 2m \log \log m + O(m), \quad (\text{S18})$$

and, for all sufficiently large  $m$ ,

$$\frac{H_{m-1}}{H_m} \leq C \left( \frac{\log m}{m} \right)^2. \quad (\text{S19})$$

For (S18), set  $r = W(2m)$ ,  $s = 2m/r$ , and evaluate the exponent at  $s$ . Since  $m/s^2 = o(1)$ ,

$$\sum_{j < m} \log(M(s) - j) = 2m \log s - m \log 2 + O(r^2).$$

Stirling's estimate,  $s \log s = 2m$ , and  $\tau - s = O(r)$  give

$$f_m(\tau) = 2m \log s - m \log 2 - 2m + s + O(r^2 + \log m).$$

Theorem S1 contributes only  $O(\log m)$  to the logarithm. Finally  $\log s = \log m - \log \log m + O(1)$ , giving (S18).

For the ratio, define the positive probability law

$$\mathbb{P}_k(X = m) = \frac{(M(m))_k}{e H_k m!}$$

on the integers with  $M(m) \geq k$ . The exact falling-factorial identity gives

$$\frac{H_{k+1}}{H_k} = \mathbb{E}_k[M(X) - k]. \quad (\text{S20})$$

The global tail estimates and the leading case of Theorem S1 show  $\mathbb{P}_k(\tau_k/2 \leq X \leq 3\tau_k/2) = 1 - O(e^{-ck})$ , after harmless polynomial factors are absorbed into the exponential. On this interval,

$$M(X) - k \geq c\tau_k^2 \geq c'k^2/(\log k)^2$$

for large  $k$ . Hence (S20) is at least this size, proving (S19) with  $m = k + 1$ . This argument does not subtract two  $O(m)$  logarithmic errors, which would not establish an adjacent-ratio estimate.

For a more precise conclusion, fix any  $0 < \varepsilon < 1/2$ . Equation (S10) makes the probability outside  $|X - \tau_k| \leq \varepsilon\tau_k$ , but inside  $[\tau_k/2, 3\tau_k/2]$ , exponentially small. Its contribution to  $\mathbb{E}_k X^2$  is also exponentially small relative to  $\tau_k^2$ . For  $X > 3\tau_k/2$ , multiply the exponential tangent majorant (S11) by  $X^2$  and sum the resulting geometric series and its first two derivatives; this bounds the second-moment tail by a polynomial in  $\tau_k$  times  $e^{-ck}$ , after normalization. The left tail is bounded similarly since  $X^2 \leq \tau_k^2$  there. Thus the central interval gives  $(1 - \varepsilon)^2 \leq \liminf \mathbb{E}_k X^2 / \tau_k^2 \leq \limsup \mathbb{E}_k X^2 / \tau_k^2 \leq (1 + \varepsilon)^2$ . Letting  $\varepsilon \downarrow 0$ , and using  $k = o(\tau_k^2)$ , (S20) gives  $H_{k+1}/H_k \sim \tau_k^2/2$ . Consequently

$$H_{n-1}/H_n \sim r^2/(2n^2). \quad (\text{S21})$$

Only the upper bound (S19) is needed below.

A useful explicitly normalized consequence is the following. Define

$$S_n = \frac{\exp\{2n \log s - n \log 2 - 2n + s - r^2/4 - r/2 - 1\}}{\sqrt{r+1}}.$$

Then

$$H_n = S_n \left(1 + O(r^4/n)\right). \quad (\text{S21a})$$

Indeed, with  $u_j = 1/s + 2j/s^2$ ,

$$\sum_{j < n} \log(1 - u_j) = -\frac{n}{s} - \frac{n(n-1)}{s^2} + O\left(\sum_{j < n} u_j^2\right) = -r/2 - r^2/4 + O(r^4/n).$$

Here  $\max u_j = O(r^2/n) = o(1)$  and  $\sum u_j^2 = O(r^4/n)$ . Positive-real Stirling expansion therefore gives

$$f_n(s) = 2n \log s - n \log 2 - 2n + s - r^2/4 - r/2 - \frac{1}{2} \log(2\pi s) + O(r^4/n).$$

The estimates  $f'_n(s) = O(r^3/n)$ ,  $\tau - s = O(r)$ , and (S8) imply  $f_n(\tau) - f_n(s) = O(r^4/n)$ . Finally combine the leading case of (S1) with the variance estimate in (S2). Since  $r^4/n \rightarrow 0$ , exponentiating the resulting logarithmic error proves (S21a).

## 2.6 Arbitrary fixed real polynomial perturbations

Let  $P(x)$  be a fixed real polynomial with  $P(0) = 0$ , and set

$$p_j = [x^j]e^{P(x)}, \quad A_P(n) = \sum_{j=0}^n (n)_j p_j H_{n-j}.$$

This is a finite coefficient identity for formal exponential generating functions. Neither  $\sum H_n x^n/n!$  nor its product by  $e^{P(x)}$  is being evaluated at positive  $x$ . Equation (S18), together with Stirling's estimate for  $n!$ , shows the former has radius of convergence zero.

**Theorem S3.** For each fixed integer  $K \geq 0$ ,

$$A_P(n) = \sum_{j=0}^K (n)_j p_j H_{n-j} + O_{P,K} \left( H_n (r^2/n)^{K+1} \right). \quad (\text{S22})$$

If  $P \neq 0$  has degree  $d \geq 1$ , the signed far tail admits the absolute bound

$$\frac{1}{H_n} \sum_{j > n/2} |(n)_j p_j H_{n-j}| \leq \exp \left[ -\frac{1+1/d}{2} n \log n + n \log \log n + O_P(n) \right]. \quad (\text{S23})$$

It also satisfies the weaker bound with leading term  $-n \log n/(2d)$ . For  $P = 0$ , the tail is identically zero.

**Proof.** For  $j \leq n/2$ , multiply (S19) over the indices  $n-j+1, \dots, n$ :

$$\frac{H_{n-j}}{H_n} \leq \left( C \frac{\log^2 n}{n^2} \right)^j.$$

Since  $(n)_j \leq n^j$ , these summands are at most  $|p_j| z_n^j$ , where  $z_n = C \log^2 n/n$ . Write  $P(x) = \sum_{\ell=1}^d c_\ell x^\ell$ . The nonnegative coefficients of  $\exp(\sum |c_\ell| x^\ell)$  majorize  $|p_j|$ . Their series is entire, so, for fixed  $K$ ,

$$\sum_{j \geq K+1} |p_j| z_n^j = O_{P,K}(z_n^{K+1}). \quad (\text{S24})$$

This bounds the entire part  $K < j \leq n/2$ .

For the far tail, Cauchy's coefficient inequality on  $|z| = j^{1/d}$  gives

$$|p_j| \leq \exp[-(j/d) \log j + O_P(j)]. \quad (\text{S25})$$

For completeness, this follows since  $\max_{|z|=R} |e^{P(z)}| \leq \exp(\sum |c_\ell| R^\ell)$ , and at  $R = j^{1/d}$  the last exponent is  $O_P(j)$ .

Equation (S18) implies the uniform upper bound

$$\log H_k \leq 2k(\log n - \log \log n) + O(n) \quad (0 \leq k \leq n), \quad (\text{S26})$$

because  $\log x - \log \log x$  is increasing for  $x > e$ , while finitely many small  $k$  can be absorbed into the constant. Its lower bound at  $n$ , together with  $(n)_j \leq n^j$ , gives, for  $k = n - j$ ,

$$\log \frac{(n)_j H_{n-j}}{H_n} \leq -j \log n + 2j \log \log n + O(n). \quad (\text{S27})$$

For  $j > n/2$ ,  $\log j \geq \log n - \log 2$ , so (S25) and (S27) bound the logarithm of the absolute normalized summand by

$$-(1 + 1/d)j \log n + 2j \log \log n + O_P(n).$$

The explicit part is decreasing in  $j$  for large  $n$ . At  $j = n/2$  it equals the exponent in (S23); summing at most  $n$  terms only adds  $O(\log n)$ . This proves (S23), which is smaller than every fixed negative power of  $n$ , and (S24) proves (S22), since  $r \asymp \log n$ .

The proof uses an absolute majorant and therefore applies unchanged when coefficients of  $P$ , or the coefficients  $p_j$ , have mixed signs. In particular (S22) with  $K = 0$  shows  $A_P(n)/H_n = 1 + O_P(r^2/n)$ , so  $A_P(n) > 0$  for all sufficiently large  $n$ , even for signed perturbations. Consequently its formal exponential generating function also has radius of convergence zero.

## 2.7 A pure power error for the perturbed exact saddle formula

For  $m \geq 1$ , let

$$\mathcal{H}_{m,K} = e^{f_m(\tau_m)-1} \sqrt{2\pi} \sigma_m \sum_{\ell=0}^K C_{m,\ell}$$

be the exact-saddle expression truncated at level  $K$ . For every fixed  $K \geq 0$  and fixed  $P$ , the preceding theorems imply

$$A_P(n) = \sum_{j=0}^{K+1} (n)_j p_j \mathcal{H}_{n-j,K} + O_{P,K}(H_n n^{-K-1}).$$

Here  $n > K + 1$ , as is automatic in the limit. Indeed, Theorem S3 with  $K + 1$  leaves  $O(H_n(r^2/n)^{K+2}) = O(H_n n^{-K-1})$ , since a fixed power of  $r$  is  $o(n)$ . On the finitely many retained shifts, Theorem S1 and the adjacent-ratio bound show that the total error of replacing  $H_{n-j}$  by  $\mathcal{H}_{n-j,K}$  is also  $O(H_n n^{-K-1})$ . This obtains a pure power error by retaining one extra fixed shift. It does not turn the same fixed Lambert truncation into a pure power error.

### 3 Lambert expansions and explicit coefficients

All limits are for positive integer  $n$  tending to infinity. The polynomial  $P$  and expansion order  $J$  are fixed before taking that limit. Implicit constants can depend on  $P$  and  $J$ . Numerical diagnostics below are not rigorous intervals.

#### 3.1 Definitions and main theorem

Write  $(u)_k = u(u-1)\cdots(u-k+1)$  for a falling factorial and  $u^{\bar{k}} = u(u+1)\cdots(u+k-1)$  for a rising factorial. Put

$$M(m) = m(m-1)/2, \quad H_n = e^{-1} \sum_{m \geq 0} \frac{(M(m))_n}{m!},$$

$$c_k = [z^k]e^{P(z)}, \quad A_P(n) = \sum_{k=0}^n (n)_k c_k H_{n-k}.$$

Assume  $P$  is a fixed real polynomial with  $P(0) = 0$ . The relevant cases are

$$P_V(z) = -z/2, \quad P_U(z) = z/2, \quad P_L(z) = z/2 - z^3/6 - z^4/4 - z^5/8 - z^6/48.$$

Let

$$r = W(2n), \quad s = e^r = 2n/r,$$

$$S_n = \frac{\exp\{2n(r-1+1/r) - 1 - n \log 2 - r^2/4 - r/2\}}{\sqrt{1+r}}. \quad (\text{C1})$$

**Theorem C1.** If  $K_P$  is the field generated over the rationals by the coefficients of  $P$ , there are rational functions  $a_{P,j}(r)$  in  $K_P(r)$ , obtained by the finite algebraic prescription below, with  $a_{P,0} = 1$  and

$$a_{P,j}(r) = O_{P,j}((1+r)^{4j}),$$

such that, for every fixed  $J \geq 0$ ,

$$A_P(n) = S_n \left( \sum_{j=0}^J \frac{a_{P,j}(r)}{n^j} + O_{P,J} \left( \frac{(1+r)^{4J+4}}{n^{J+1}} \right) \right). \quad (\text{C2})$$

For rational  $P$  these are ordinary rational functions over the rationals and the prescription is exactly computable by rational arithmetic. For arbitrary real  $P$  the prescription is algebraic relative to its coefficients; no computable access to an arbitrary real constant is assumed.

Thus the statement covers every fixed order. The accompanying software has been run and checked through  $J = 3$ ; computation at three orders is not the proof of the all-orders assertion. The public interface supports orders zero through three. The finite algebraic prescription itself has no fixed mathematical order cap.

### 3.2 Exact finite-order coefficient generator

Use formal  $h$  and  $y$ , with

$$h = s^{-1/2}, \quad x = 1 + hy, \quad n = r/(2h^2), \quad m = x/h^2, \quad M = \frac{x^2}{2h^4}(1 - h^2/x).$$

Write  $B_k(z)$  for the Bernoulli polynomial and  $B_k = B_k(0)$  for the Bernoulli number. These differ from the Bell numbers  $B_k$  used earlier; for example  $B_2 = 1/6$ . For  $p \geq 1$  let

$$F_p(n) = \sum_{j=0}^{n-1} j^p = \frac{B_{p+1}(n) - B_{p+1}(0)}{p+1}.$$

The identity in the second expression defines its polynomial continuation. Set

$$D = n \log(1 - h^2/x) - \sum_{p \geq 1} \frac{F_p(n)}{pM^p}, \quad (C3)$$

$$E = \frac{(r-x) \log x + (1-r)(x-1)}{h^2} + \frac{r+1}{2}y^2 - \frac{1}{2} \log x - \sum_{q \geq 1} \frac{B_{2q}h^{4q-2}}{2q(2q-1)x^{2q-1}} + D + \frac{r}{2} + \frac{r^2}{4}, \quad (C4)$$

$$Q = \sum_{k \geq 0} c_k \frac{(n)_k}{(M-n+1)^{\bar{k}}}. \quad (C5)$$

Equations (C3)–(C5) are finite-order formal instructions. To obtain  $a_{P,j}$ , keep  $p \leq j+1$  in (C3),  $q$  with  $4q-2 \leq 2j$  in (C4), and  $k \leq j$  in (C5), and discard powers above  $h^{2j}$ . In particular, no infinite Stirling series is asserted to converge.

If  $Y$  is centered Gaussian with variance  $1/(r+1)$ , then

$$a_{P,j}(r) = \left(\frac{r}{2}\right)^j \mathbb{E} \left( [h^{2j}] e^{E(h,Y,r)} Q(h,Y,r) \right). \quad (C6)$$

This is an exact algebraic prescription over  $K_P$ , and is an exact rational-arithmetic prescription when  $P$  has rational coefficients. Gaussian moments are

$$\mathbb{E}Y^{2\ell} = (2\ell-1)!!/(1+r)^\ell, \quad \mathbb{E}Y^{2\ell+1} = 0.$$

An efficient implementation computes the coefficients of  $\exp(E)$  by the finite recurrence

$$b_0 = 1, \quad b_j = \frac{1}{j} \sum_{k=1}^j k E_k b_{j-k}, \quad E = \sum_{k \geq 1} E_k h^k.$$

The code `generate_coefficients.py` implements this using exact SymPy rational arithmetic for the displayed rational polynomials  $P$ ; formal polynomial parameters can instead be retained symbolically. Setting  $D = 0$  and omitting its centering constants gives the usual Bell saddle corrections  $b_j$  for

$$2^{-n} B_{2n} = \frac{e^{2n(r-1+1/r)-1-n \log 2}}{\sqrt{1+r}} \left( \sum_{j=0}^J b_j(r) n^{-j} + O_J((1+r)^{4J+4} n^{-J-1}) \right).$$

The coefficients relative to the exact Bell factor are obtained by ordinary formal division, not by identifying  $S_n$  with that exact factor.

### 3.3 Justification of the fixed-order remainder

#### 3.3.1 Localization and signed convolution

The preceding exact-saddle section establishes the following facts directly from strict concavity of

$$f_n(t) = \sum_{j < n} \log(M(t) - j) - \log \Gamma(t + 1), \quad t > (1 + \sqrt{1 + 8(n-1)})/2.$$

Its unique saddle  $\tau$  satisfies  $\tau = s + O(r)$ , its variance is  $\sigma^2 = s(1 + O(r^2/n))/(r+1)$ , and on an interval comparable with  $s$  the exponent loses at least  $c(t - \tau)^2/\sigma^2$ . Outside that interval a concave tangent gives an exponentially decreasing right tail; the left tail has only  $O(s)$  integer points. In particular  $H_n/S_n = 1 + O(r^4/n)$ . The same estimates hold with  $n$  replaced by  $n - k$  for any fixed  $k$ .

Positivity also gives the adjacent-ratio upper bound

$$H_{n-1}/H_n \leq C(\log n/n)^2.$$

Combining it for  $k \leq n/2$  with the entire majorant  $\exp(\sum_j |[z^j]P|z^j)$ , and using a coefficient Cauchy bound for  $k > n/2$ , yields

$$A_P(n) = \sum_{k=0}^J (n)_k c_k H_{n-k} + O_{P,J}(H_n(r^2/n)^{J+1}). \quad (C7)$$

This is an absolute bound and does not assume the  $c_k$  have one sign. For  $d = \deg P \geq 1$  the normalized absolute far tail  $k > n/2$  is at most

$$\exp\{-\frac{1}{2}(1 + 1/d)n \log n + n \log \log n + O_P(n)\}.$$

Therefore only fixed  $k \leq J$  are needed for the Lambert coefficient calculation.

For each such  $k$ , discard its extra left support  $n - k \leq M(m) < n$ . There  $m = O(\sqrt{n})$ , and the logarithm of its unnormalized summand is at most  $n \log n + O_J(n + \log n)$ , whereas  $\log H_n = 2n \log n - 2n \log \log n + O(n)$ . It is exponentially negligible relative to every fixed algebraic error required here. On the common support  $M(m) \geq n$ , the exact identity

$$\frac{(M)_{n-k}}{(M)_n} = \frac{1}{(M - n + 1)^k} \quad (C8)$$

justifies the finite version  $Q_J$  of (C5). Applying (C8) to the discarded support would be invalid.

Nor is a uniform bound on  $Q_J$  near  $M = n$  needed. Localize each original positive sum  $H_{n-k}$  separately before introducing  $Q_J$ . For fixed  $k$  its saddle differs from  $s$  by  $O_J(r)$ , and its variance is comparable with  $s/(1+r)$ . Thus the portion outside the common Lambert window  $|Z| \leq r^2$  is  $O_J(H_{n-k}e^{-c_J r^4})$ . Multiplying by  $(n)_k |c_k|$  and using  $(n)_k H_{n-k}/H_n = O_J((r^2/n)^k)$  gives  $O_{P,J}(H_n e^{-c_J r^4})$  after summing  $k \leq J$ . This controls the boundary and far tails even when the rational multiplier itself is large there. Only inside the common central window is  $Q_J$  expanded analytically.

#### 3.3.2 A finite analytic exponent

Fix  $J$ . In (C3), retain  $p = 1, \dots, J+1$  and retain the logarithm  $n \log(1 - h^2/x)$  analytically, with its removable value at  $h = 0$ . In (C4), retain only those Stirling terms with  $4q - 2 \leq 2J$ . Call the resulting finite analytic exponent  $E_J$ . Use  $Q_J$  from (C5) with  $k \leq J$ .

The original real collision logarithm differs from this finite version, uniformly for  $m$  comparable with  $s$ , by

$$O_J(n^{J+3}/M^{J+2}) = O_J(r^{J+3}h^{2J+2}). \quad (\text{C9})$$

Indeed the omitted logarithm tail is bounded by a constant times  $\sum_{j < n} (j/M)^{J+2}$ , since  $n/M = o(1)$ . The omitted real positive-argument Stirling remainder is bounded by its first omitted term, and hence is  $O_J(h^{2J+2})$  or smaller. These are estimates for real  $h$ ; they are not deduced from convergence of the Stirling series. The standard finite Stirling remainder is documented in DLMF §5.11.

At  $h = 0$ ,  $D$  begins with  $-r/(2x) - r^2/(4x^2)$ , so  $E_J(0, y, r) = 0$ . Every singular-looking power of  $h$  in the finite expression is removable. For example,  $F_p(n)/M^p$  starts at  $h^{2p-2}$ ; the quadratic term of the first expression in (C4) cancels its explicitly added Gaussian term.

### 3.3.3 Cauchy radius, integrated remainder, and parity

Rescale  $Z = \sqrt{1+r} y$ . For real  $Z$  and  $r \geq 1$ , there is a sufficiently small  $c_J > 0$  such that on the complex  $h$ -disk

$$|h| \leq R_Z = c_J(1+r)^{-3/2}(1+|Z|)^{-3} \quad (\text{C10})$$

we have

$$|e^{E_J(h, Z/\sqrt{1+r}, r)} Q_J(h, Z/\sqrt{1+r}, r)| \leq C_{P,J}. \quad (\text{C11})$$

Here is a direct bound behind (C11). On this disk,  $|hy| \leq c_J(1+r)^{-2}|Z|/(1+|Z|)^3$ , so all denominators  $x$  and the finitely many factors in  $Q_J$  stay away from zero. The shifted leading collision part is  $O(r^2|hy|)$ , hence bounded. The next collision contribution is  $O(r^3|h|^2)$ , hence bounded, and all subsequent fixed collision terms are smaller. The cubic-and-higher Bell phase is  $O((1+r)|h||y|^3/(1-|hy|))$ , also bounded. The logarithmic amplitude and the finite Stirling terms are bounded as well. The finite  $Q_J$  is bounded since each  $k$ -term starts at  $O((r|h|^2)^k)$  and all its normalized denominators are nonzero. Constants may be reduced or enlarged for finitely many lower powers and fixed coefficients.

Cauchy's estimate therefore bounds its coefficient of  $h^q$  by

$$C_{P,J,q}(1+r)^{3q/2}(1+|Z|)^{3q}. \quad (\text{C12})$$

On the real central window  $|Z| \leq r^2$ , the actual  $h = e^{-r/2}$  eventually satisfies  $h \leq R_Z/2$ . Expanding through degree  $2J+1$  has pointwise remainder bounded by

$$C_{P,J}h^{2J+2}(1+r)^{3J+3}(1+|Z|)^{6J+6}. \quad (\text{C13})$$

Multiplying by the Gaussian density and integrating gives

$$O_{P,J}(h^{2J+2}(1+r)^{3J+3}) = O_{P,J}((1+r)^{4J+4}/n^{J+1}). \quad (\text{C14})$$

The real errors in (C9), and the Stirling error, fit inside (C14); exponentiation preserves those bounds because they tend uniformly to zero on this window. The convolution remainder (C7) also fits, using  $H_n/S_n = 1 + o(1)$ .

For clarity about the lattice:  $Z_m = \sqrt{1+r} h(m-s)$ , so its mesh is  $\Delta = \sqrt{1+r} h$ . The normalized central sum is

$$\frac{\Delta}{\sqrt{2\pi}} \sum_m e^{-Z_m^2/2} e^{E_J(h, Z_m/\sqrt{1+r}, r)} Q_J(h, Z_m/\sqrt{1+r}, r). \quad (\text{C15})$$

The mesh-weighted sum of any fixed polynomial times  $e^{-Z^2/2}$  is uniformly bounded. Thus (C13) controls the summed remainder too. For each of the finitely many retained polynomial coefficients, Poisson summation of a Gaussian and its derivatives replaces (C15) by its integral with error bounded by a polynomial in  $r$  and  $\Delta^{-1}$  times  $\exp(-2\pi^2/\Delta^2)$ , hence smaller than every fixed required power of  $n$ . The actual sum outside  $|Z| \leq r^2$  has relative size  $\exp(-cr^4)$ , by the concavity estimates and  $\tau - s = O(r)$ ; fixed polynomial factors do not change its negligibility. This also justifies extending the polynomial-Gaussian integrals to the whole line.

Finally the analytic expression is invariant under  $(h, y) \mapsto (-h, -y)$ . Its coefficient of  $h^q$  has parity  $q$  in  $y$ . Odd terms integrate to zero. The even terms are precisely (C6). Formula (C12), integrated at  $q = 2j$  and multiplied by  $(r/2)^j$ , gives  $a_{P,j} = O((1+r)^{4j})$ . Gaussian moments put the coefficients in  $K_P(r)$ , and in  $\mathbb{Q}(r)$  for rational  $P$ . This proves (C2).

### 3.4 Explicit coefficients and the nonconstant remainder issue

The  $S_n$ -normalized first coefficients are

$$\begin{aligned} a_{H,1} &= \frac{r^2(r^5 - 12r^3 - 28r^2 - 28r - 16)}{48(r+1)^3}, \\ a_{V,1} &= \frac{r^2(r^5 - 24r^3 - 64r^2 - 64r - 28)}{48(r+1)^3}, \\ a_{U,1} &= a_{L,1} = \frac{r^2(r^5 + 8r^2 + 8r - 4)}{48(r+1)^3}. \end{aligned}$$

For compact display, define the numerator polynomials

$$\begin{aligned} N_H(r) &= r^{11} - 48r^9 - 80r^8 + 544r^7 + 2032r^6 + 2176r^5 \\ &\quad - 544r^4 - 2928r^3 - 2368r^2 - 1088r - 768, \\ N_V(r) &= r^{11} - 72r^9 - 152r^8 + 904r^7 + 3832r^6 + 3280r^5 \\ &\quad - 7936r^4 - 22272r^3 - 23296r^2 - 12080r - 3072, \\ N_U(r) &= r^{11} - 24r^9 - 8r^8 + 472r^7 + 1960r^6 + 5392r^5 \\ &\quad + 12608r^4 + 20736r^3 + 20288r^2 + 10192r + 1536. \end{aligned}$$

The second coefficients are

$$a_{H,2} = \frac{r^3 N_H(r)}{4608(r+1)^6}, \quad a_{V,2} = \frac{r^3 N_V(r)}{4608(r+1)^6}, \quad a_{U,2} = a_{L,2} = \frac{r^3 N_U(r)}{4608(r+1)^6}.$$

The equality of  $U$  and  $L$  at these two orders is structural:  $e^{P_L}$  and  $e^{P_U}$  agree through degree two. At the next order,

$$a_{L,3} - a_{U,3} = -r^6/48 \quad (\text{C16})$$

exactly. The complete order-three rational functions are in `data/coefficients_order3.json`. Now define the **exact** Bell normalization

$$T_n = 2^{-n} B_{2n} e^{-r^2/4 - r/2}.$$

Then

$$\frac{V_n}{T_n} = 1 + \frac{v_1(r)}{n} + \frac{v_2(r)}{n^2} + \frac{v_3(r)}{n^3} + O(r^{16}/n^4),$$

where

$$v_1(r) = r^2(r-6)(r+3)/48,$$

$$\begin{aligned} v_2(r) &= \frac{r^3 N_B(r)}{4608(r+1)^3}, \\ N_B(r) &= r^8 - 3r^7 - 66r^6 + 58r^5 + 945r^4 + 813r^3 \\ &\quad - 2640r^2 - 4956r - 2208. \end{aligned}$$

and

$$v_3(r) \sim r^{12}/663552.$$

Consequently the remainder after  $v_2/n^2$  is asymptotic to  $r^{12}/(663552n^3)$ . It is **not**  $O(n^{-3})$  **with a constant independent of  $n$** . A valid simple upper bound at that truncation is  $O(r^{12}/n^3)$ .

[write] **How slowly  $v_3$  reaches its leading term, 7 October 2026.** The statements  $v_3(r) \sim r^{12}/663552$  and the resulting asymptotic form of the remainder are correct (the write's derivation agrees, and  $v_3/r^{12} \rightarrow 1/663552$ ), but the leading term takes over only at astronomically large  $n$ . The numerator of  $v_3$  has its largest positive root at  $r = 10.1555\dots$ :  $v_3(W(2n)) < 0$  for  $2190 \leq n \leq 130671$  ( $6.5111\dots < r < 10.1555\dots$ ), and  $v_3/(r^{12}/663552)$  is  $-0.0517, -0.0065, 0.0885, 0.277$  at  $r = 8, 10, 12, 16$ , reaching  $1/2$  only at  $r = 23.09\dots$  ( $n \approx 1.2 \times 10^{11}$ ) and  $0.9$  at  $r = 99.08\dots$  ( $n \approx 5 \times 10^{44}$ ). Numerically, with the full  $v_3$  the remainder after  $v_2/n^2$  is matched to relative accuracy  $10^{-3}$ – $10^{-5}$  for  $n = 10^4, \dots, 10^7$  (Section 1.5). The bound  $O(r^{12}/n^3)$  stated by the source is the safe reading.

The first relative effect of identifying line images is consequently

$$\frac{L_n}{U_n} = 1 - \frac{r^6}{48n^3} + O\left(\frac{r^{16}}{n^4}\right).$$

This follows by dividing the two expansions with their common first and second coefficients. The correction separating the legacy series from corrected line graphs first enters at order four, consistently with the missing quartic term of the legacy multiplier.

## 4 Inversion and threshold certification

### 4.1 Inversion on the sequence range

Let  $Y$  denote the target value. Define

$$\Phi(x) = 2x(r-1+1/r) - x \log 2, \quad r = W(2x),$$

$$h_0(r) = r^2/4 + r/2 + 1 + \frac{1}{2} \log(1+r).$$

Direct differentiation using  $r' = r/[x(1+r)]$  gives

$$\Phi'(x) = 2r - \log 2 =: q(r), \quad \Phi''(x) = \frac{2r}{x(1+r)}. \quad (\text{C17})$$

From (C2),

$$\log A_P(n) = \Phi(n) - h_0(r) + a_{P,1}(r)/n + O_P(r^8/n^2). \quad (\text{C18})$$

For large  $Y$ , let  $x_0$  be the unique large real solution of  $\Phi(x_0) = \log Y$  and let  $r_0 = W(2x_0)$ . If  $Y = A_P(N)$  for an integer  $N$  tending to infinity, then

$$N = x_0 + \frac{h_0(r_0)}{2r_0 - \log 2} + O_P(r_0^3/x_0). \quad (\text{C19})$$

This is an absolute index error, not a relative one. To see the scale, the leading displacement is  $O(r_0)$ ; inserting that displacement in (C18) leaves  $O(r_0^4/x_0)$ , which is divided by  $\Phi'$  asymptotic to  $2r_0$ .

One additional explicit correction is available. Write

$$d_0(r) = h_0(r)/q(r), \quad d_1(r) = \frac{\frac{r}{1+r}h'_0(r)d_0(r) - \frac{r}{1+r}d_0(r)^2 - a_{P,1}(r)}{q(r)}.$$

A Taylor expansion of (C18), now retaining its  $O(r^8/n^2)$  error, gives

$$N = x_0 + d_0(r_0) + d_1(r_0)/x_0 + O_P(r_0^7/x_0^2). \quad (\text{C20})$$

For details of the second correction, let  $x = x_0$ ,  $r = W(2x)$  and  $G(t) = \Phi(t) - h_0(W(2t)) + a_{P,1}(W(2t))/t$ . For a displacement  $d = O(r)$ , finite Taylor expansion gives

$$G(x+d) - \log Y = q(r)d - h_0(r) + \frac{\frac{r}{1+r}d^2 - \frac{r}{1+r}h'_0(r)d + a_{P,1}(r)}{x} + O_P(r^8/x^2).$$

Indeed  $a_{P,1} = O(r^4)$  and its rational derivative is  $O(r^3)$ ; all higher Taylor terms are smaller than the displayed error. Taking  $d = d_0 + d_1/x$  cancels the constant and  $x^{-1}$  terms. Equation (C18) at the true integer  $N$ , followed by the mean value theorem for the explicit  $G$  with  $G'(t) \sim 2W(2t)$ , proves (C20). The preliminary localization  $N = x_0 + O(r_0)$  follows first from (C18) and  $\Phi'(x) \asymp r_0$  in a fixed relative neighborhood of  $x_0$ . No smooth continuation of the unknown error is used.

For arbitrary fixed  $J$ , set

$$F_J(x) = \Phi(x) - h_0(W(2x)) + \log \left( \sum_{j=0}^J a_{P,j}(W(2x))/x^j \right). \quad (\text{C21})$$

The logarithm has positive argument eventually, and direct differentiation of the finite rational formula shows  $F'_J(x) \geq W(2x)$  eventually. If  $x_J$  is its large root  $F_J(x_J) = \log Y$  and  $Y = A_P(N)$ , then

$$|x_J - N| = O_{P,J} \left( (1+r_N)^{4J+3}/N^{J+1} \right). \quad (\text{C22})$$

No differentiation of the unknown asymptotic remainder is used: apply (C2) only at the integer  $N$ , and apply the mean value theorem to the explicit smooth  $F_J$ .

## 4.2 Lambert seed and its limitation

Set

$$a = 1 + \frac{1}{2} \log 2, \quad b = 1 - a^2/4,$$

$$w = 2W\left(\frac{1}{2}\sqrt{\log Y} e^{-a/4}\right).$$

The equation  $\Phi(x_0) = \log Y$  is exactly

$$e^{r_0}(r_0^2 - ar_0 + 1) = \log Y.$$

Writing  $r_0 = w + a/2 + \delta$  reduces it to

$$e^\delta((w + \delta)^2 + b) = w^2,$$

so

$$\delta = -\frac{b}{w(w + 2)} + O(w^{-4}). \quad (\text{C23})$$

This provides a useful initial seed for solving the exact scalar equation for  $r_0$ , then  $x_0 = r_0 e^{r_0}/2$ . The defining Lambert relation and branch convention are standard; see DLMF §4.13.

A fixed finite truncation of this  $1/w$  seed does **not** retain the fine absolute accuracy in (C19), (C20), or (C22). An  $r_0$  error  $O(w^{-K})$  produces an  $x_0$  error  $O(x_0 w^{-K})$ , enormously larger than the desired  $O(r_0^3/x_0)$ . Refine the exact scalar equation numerically to the needed precision, or keep the seed error explicitly in the final error bound.

## 4.3 Arbitrary thresholds require a true-sequence enclosure

The asymptotic  $A_P(n)/H_n \rightarrow 1$  and  $H_{n+1}/H_n \rightarrow \infty$  imply  $A_P(n)$  is positive and strictly increasing eventually. For large targets exceeding the maximum of the finite exceptional prefix, let  $N_Y$  be the least integer with  $A_P(N_Y) \geq Y$ . If  $n_*$  is an index beyond which positivity and monotonicity hold, this least index lies in the increasing tail once  $Y > \max_{0 \leq k < n_*} A_P(k)$ . Equation (C22), as stated, concerns targets on the sequence range. For arbitrary  $Y$ , simply rounding a root of  $F_J$  does not automatically certify  $N_Y$ .

After the exceptional prefix has been excluded, a rigorous threshold bracket requires either adjacent exact counts in the increasing tail, or a proved bound

$$|\log A_P(k) - F_J(k)| \leq E_J(k)$$

with an explicit applicable constant and starting index. Integers  $k_-$  and  $k_+$ , both in that certified increasing tail, then certify

$$F_J(k_-) + E_J(k_-) < \log Y, \quad F_J(k_+) - E_J(k_+) \geq \log Y$$

and hence  $k_- < N_Y \leq k_+$ . A numerical bracket for the model root alone does not provide those inequalities for the true sequence. The theorem's existence of an  $O$ -constant is not an implemented certified error bound.

For  $V, U, L$  within the public exact range, a finite scan checks all earlier indices instead of relying on an unimplemented monotonicity onset. Outside that range, the present asymptotic theorem alone supplies no effective certificate for a particular arbitrary threshold.

*Remark 1* ([write] The inverse and the transseries volume, 7 October 2026). Compared statement by statement with the repository’s transseries volume [7]; the volume’s symbols are not used here.

1. *The growth.*  $\log A_P(n) = 2n \log n - 2n \log \log n + O(n)$  (S18), with a term  $n \log n$ , so the sequences are not of the exponential–power form of `p0:def:model`; no instance of the volume’s model is claimed.
2. *The model root and the Lambert seed.* In the variable  $r_0 = W(2x_0)$  the model equation  $\Phi(x_0) = \log Y$  is exactly  $e^{r_0}(r_0^2 - ar_0 + 1) = \log Y$  (checked symbolically), that is  $r_0 + 2 \log r_0 + \log(1 - a/r_0 + 1/r_0^2) = \log \log Y$ . This is literally the volume’s monomial–logarithmic equation (`plt:def:lw-monomial-log-datum`) in the unknown  $r_0$ , with unit slope, logarithmic coefficient 2, zero constant and a perturbation  $\log(1 - at + t^2)$ ,  $t = 1/r_0$ , of positive order with constant coefficients; the three hypotheses of `plt:thm:lw-template` hold, so the formal expansion of  $r_0$  in the reciprocal of  $\log \log Y$  is an exact instance of the template. Its dominant block is solved exactly by the seed:  $w = 2W(\frac{1}{2}\sqrt{\log Y} e^{-a/4})$  is the solution of  $w + 2 \log w = \log \log Y - a/2$ , an exact instance of `p0:thm:lambert-core` (unit slope, positive logarithmic coefficient 2, principal branch), and with  $r_0 = w + a/2 + \delta$  the expansion (C23) is the first term of the remaining correction, a series in  $1/w$  with constant coefficients.
3. *The index.* The refinements (C19), (C20) and (C22) move  $x_0$  by  $O(r_0)$ , a relative change of order  $r_0 e^{-r_0}$ : below every order of the template’s chart. So they are not instances of `plt:thm:lw-template`; they are proved directly, at the integer  $N$ , by the mean value theorem for the explicit models.
4. *The thresholds.*  $N_Y$  is the volume’s staircase (`p0:def:three-inverses(1)`) of  $A_P$  on its increasing tail. The models  $\Phi - h_0$  and  $F_J$  are smooth and increasing but do not take the values  $\log A_P(k)$  at the integers, so they are not admissible interpolations: reading (C22) as  $N = \lfloor x_J + \frac{1}{2} \rfloor$  for large  $N$  is an analogue of `p0:thm:staircase(3)`, and the bracket  $k_- < N_Y \leq k_+$  of Section 4.3, proved directly at the integers from a true-sequence enclosure, is an analogue of `p0:thm:staircase(2)`, not an instance; the source’s caution that rounding a model root certifies nothing is the volume’s separation caveat.

## 5 Exact computation and reproducibility

### 5.1 Three distinct levels of evidence

The analytic theorems establish the fixed-order limits. Exact algebra verifies the displayed coefficient identities, while finite enumeration checks the models and indexing. High-precision floating-point residuals illustrate convergence but are not interval certificates. In particular, none of the decimal diagnostics supplies an effective constant or starting index for the asymptotic error bounds.

The standard-library program `code/exact_counts.py` implements equation (3). It builds Bell numbers with the Bell triangle, multiplies the integer polynomials recursively, and applies each exponential multiplier with exact rational arithmetic. For  $p_j = [x^j]e^{P(x)}$ , it uses

$$p_0 = 1, \quad p_j = \frac{1}{j} \sum_{k=1}^j k[x^k]P(x) p_{j-k}.$$

The resulting coefficient identity is the finite convolution from Theorem S3. Integer outputs for  $V, U, L, O$  are checked explicitly. The public exact-computation range is  $0 \leq n \leq 640$ . This is a

software limit; the theorems have no index cap. Local chunked decimal conversion avoids changing Python’s process-wide integer-string safety limit.

`code/test_exact.py` compares all five models with an independently generated exact fixture through index 80, all downloaded terms of A060053 and A014500 through 100, and all 18 terms of the legacy A132219 series. It also verifies  $U_n = \sum_{k=0}^n \binom{n}{k} V_k$  across the entire requested computation range. Failures use explicit exceptions rather than Python assertions, so the tests remain active under optimized Python.

`code/endpoint_check.py` independently enumerates set partitions of the  $2n$  labelled endpoints. It rejects loop and repeated-edge partitions; the sorted incident-label blocks canonically identify roots, and shared endpoints determine their labelled line graphs. The fixture through  $n = 6$  was freshly reproduced. The default build reruns through  $n = 5$ ; the more expensive  $n = 6$  check is available separately. This approach does not use the corrected line-graph multiplier to generate its objects.

## 5.2 Symbolic coefficients and numerical diagnostics

The optional SymPy program `code/generate_coefficients.py` directly implements (C3)–(C6). Its supported and tested order range is  $0 \leq J \leq 3$ . Frozen outputs include all four model coefficient arrays, the Bell array and the exact-Bell normalized  $V$  array through order three. The separate verifier checks the two displayed  $v_j$ , the leading term of  $v_3$ , the third-order line-graph separation, the cross-order identities and an independent first-order Gaussian calculation.

The latter calculation is short enough to state. Writing  $E = hE_1 + h^2E_2 + \dots$  gives

$$E_1 = \frac{2r+1}{6}y^3 + \frac{r^2+r-1}{2}y,$$

$$E_2 = -\frac{3r+1}{12}y^4 + \left(\frac{1}{4} - \frac{r}{2} - \frac{3r^2}{4}\right)y^2 - \frac{1}{12} + \frac{r}{4} - \frac{r^2}{4} - \frac{r^3}{12}.$$

For  $p_1 = [z]P(z)$ , the coefficient is

$$a_{P,1} = \frac{r}{2}\mathbb{E}\left(E_2 + \frac{1}{2}E_1^2 + p_1r\right), \quad \mathbb{E}y^{2j} = \frac{(2j-1)!!}{(1+r)^j}.$$

This independently reproduces the first coefficients for  $H, V, U, L$ .

The optional mpmath program `code/check_saddle.py` solves the strictly concave saddle equation and evaluates the positive defining sum using log-gamma expressions. Concavity gives a geometric upper-tail formula, which is evaluated in floating point. That evaluated bound, including its roundoff, is a diagnostic and is not advertised as a rigorous interval. The supported parameters are  $2 \leq n \leq 10000$ ,  $0 \leq K \leq 3$ , and 40 through 200 decimal digits. The lower bound  $n = 2$  ensures that the adjacent positive sum also has positive index.

At  $n = 1000$  the observed relative errors, in the convention approximation divided by the defining sum minus one, are

| $K$ | relative error            | $n^{K+1}$ times error |
|-----|---------------------------|-----------------------|
| 0   | $-1.46915 \cdot 10^{-5}$  | $-0.0146915$          |
| 1   | $-1.17884 \cdot 10^{-9}$  | $-0.00117884$         |
| 2   | $+6.29179 \cdot 10^{-13}$ | $+0.000629179$        |
| 3   | $+2.36925 \cdot 10^{-16}$ | $+0.000236925$        |

The normalized ratio  $(H_{n-1}/H_n)/(r^2/n^2)$  is about 0.501922, consistent with the proved limiting value  $1/2$ . These observations do not replace the uniform derivative, lattice or tail estimates.

[write] **The table recomputed, 7 October 2026.** The write recomputed the table at  $n = 1000$  with its own code ( $\tau = 345.4941688051\dots$ ,  $\sigma = 7.0490293656\dots$ ,  $C_1 = 1.4690496 \times 10^{-5}$ ,  $C_2 = 1.1794892 \times 10^{-9}$ ,  $C_3 = -6.2895152 \times 10^{-13}$ ): every printed value is the correct rounding of the recomputed one and of `data/saddle_diagnostics.json`. The printed values that are not truncations are  $-1.46915 \cdot 10^{-5}$  and  $-0.0146915$  ( $K = 0$ ; truncations  $-1.46914 \cdot 10^{-5}$  and  $-0.0146914$ , from  $-1.4691459\dots \cdot 10^{-5}$ ) and the ratio 0.501922 (truncation 0.501921, from 0.5019216358...). At  $n = 20$  the same normalized errors are 0.0154, 0.0051, 0.0026, 0.0035: the fixed-order behaviour is visible only for larger  $n$ .

### 5.3 Rebuilding and integer inverse queries

The accompanying README gives complete commands. The exact tests need Python 3.11 or later and no third-party library. Building the PDF additionally needs a TeX Live installation with the packages loaded in `src/report224.tex`. SymPy and mpmath are optional. The build uses a private TeX format, fixed PDF metadata, a sorted uncompressed ZIP with fixed timestamps, and an output directory outside the source tree. It verifies a source manifest before running and checks that no source bytes changed. Byte reproducibility is claimed for the same toolchain, not across arbitrary TeX or library versions.

The exact threshold command scans a finite computed sequence in order and returns its first value at least the specified positive integer threshold. It does not assume monotonicity at small indices. If no crossing occurs, it says only that no crossing was found in the checked range. Threshold input and output use bounded chunked decimal conversion. It must not be described as an unbounded exact inverse oracle.

A full package reproduction consists of ordinary and optimized cap-640 builds, comparison of the actual member names and bytes and the entire ZIP, plus visual inspection of every PDF page. A matching ZIP certifies reproducible bytes, not mathematical truth; the proof review and visual checks remain separate.

[write] **The shipped files, 7 October 2026.** In ProveIt the PDF and the manifest are not shipped; the text is `article.tex` with `sections/*.tex`, the programs are `code/*.py` (the builder as `code/build.py`), the data, the build receipt and the retrieved b-files are under `data/`. The tests read the b-files from `sources/` and the builder verifies the delivered manifest, so they run only in a re-extracted archive; the report README gives commands for that. The “packages loaded in `src/report224.tex`” are those of `article.tex`; the write added none.

*Remark 2* ([write] The OEIS entries, 7 October 2026). The write read the entries in the internal format on 7 October 2026. *A060053*, revision #51 (30 May 2026), by Vladeta Jovovic (15 February 2001): “Number of r-bicoverings (or restricted proper 2-covers) of an n-set.”, with links to the arXiv preprint and the journal article of Cameron, Prellberg and Stark (“See  $v_n$ ”), the relation  $U(x) = e^x V(x)$ , an e.g.f. by Andrew Howroyd (2020) and a b-file by Robert Gerbicz for  $n = 0, \dots, 100$ . *A014500*, revision #52 (30 May 2026), by Simon Plouffe and Gilbert Labelle: “Number of graphs with unlabeled (non-isolated) nodes and n labeled edges.”, with the journal link (“see  $u_n$ ”) and a b-file by Alois P. Heinz for  $n = 0, \dots, 100$ . Neither entry has an asymptotic formula or a conjecture, as the source states. *A132219*, revision #5 (7 July 2015), by Vladeta Jovovic (14 August 2007): “Number of line graphs on n labeled nodes.”, with the e.g.f.  $\exp(x/2 - x^3/6) \cdot \text{Sum}(A020556(n) \cdot (\log(1+x)/2)^n / n!, n=0..infinity)$ , that is  $e^{-x^3/6} U(x)$ , the legacy series  $O$ , and 18 displayed terms (offset 1; the served b-file lists these terms). Its terms do not match its name:  $a(4) = 66$  exceeds the  $2^6 = 64$  labelled graphs on four vertices, and the true count is  $L_4 = 60$ ; its formula is Proposition 5 of the arXiv version 1, which the corrected

manuscript supersedes. The corrected sequence 1, 1, 2, 8, 60, 729, 11600, 228443, ... is not in the OEIS (search of 7 October 2026). *A094089*, revision #13 (18 November 2020), by Vladeta Jovovic (1 May 2004), “E.g.f.:  $\exp(-1) \cdot \text{Sum}((1+2^*x)^{\text{binomial}(n,2)}/n!, n=0..\text{infinity})$ .”, b-file by Heinz for  $n = 0, \dots, 150$ : it is  $2^n H_n$ , a sequence the source tabulates without naming. *Checks*. All 101 b-file terms of A060053 and A014500, the 18 terms of A132219 and all 151 terms of A094089 equal the write’s computation; the retrieved b-files shipped in **data/** equal the live ones byte for byte. No OEIS edit is part of this work.

## 6 Further research questions

1. **Effective enclosures.** Can the strict-concavity and derivative bounds be made explicit enough to give practical constants and starting indices for each fixed order? This would turn model-root estimates into certified threshold searches beyond the exact software cap.
2. **Growing orders.** The present constants depend on a fixed order. What is the growth of the exact-saddle contraction coefficients and of the Lambert rational functions as the order grows? An optimal-truncation theorem requires information absent from a fixed-order expansion.
3. **Exponential lattice corrections.** The lattice error is smaller than every fixed algebraic order. Can its leading oscillatory dependence on the noninteger saddle be extracted, and how does it compare with other exponentially small corrections?
4. **Further graph observables.** Marking finitely many connected components leads to multivariate polynomial perturbations. A useful extension would give joint local limits or conditioned expansions while retaining the signed-tail controls used here.
5. **Uniform inverse computation.** Can an interval implementation of the positive sum, exact-saddle derivatives and inverse correction bridge small exact counts to asymptotic enclosures with an explicit guarantee? Floating-point residuals alone do not settle this.
6. **Source history and coverage.** The obsolete line-graph multiplier remains in the retrieved A132219 data, while the corrected author manuscript is clear. A comprehensive bibliographic study could determine which higher-order formulas or inverse techniques have appeared elsewhere. This report makes no inference of novelty from the absence of such formulas in the checked sources.

[write] **Added 7 October 2026, under Vladimir’s standing rule of 4 October 2026.** These six questions are the source’s own and stay open. The write adds from the non-claims: an independent check of the completeness of the exception classification (the write’s enumeration reaches seven edges), interval certificates for the diagnostics, and the correction of the OEIS entry A132219, which is not part of this work. No claim of the source was found false.

## References

- [1] P. J. Cameron, T. Prellberg and D. Stark, *Asymptotic enumeration of 2-covers and line graphs*, Discrete Mathematics **310** (2010), no. 2, 230–240. doi:10.1016/j.disc.2008.09.008. Corrected 20-page author manuscript: <https://webSPACE.maths.qmul.ac.uk/p.j.cameron/preprints/cover.pdf>.

- [2] P. J. Cameron, T. Prellberg and D. Stark, original 19-page preprint, 4 July 2007, <https://arxiv.org/pdf/0707.0664v1>. The Proposition 5 formula in this version is superseded by [1].
- [3] OEIS Foundation Inc., The On-Line Encyclopedia of Integer Sequences, A060053, restricted proper two-covers. <https://oeis.org/A060053>. Entry and b-file retrieved 5 October 2026.
- [4] OEIS Foundation Inc., The On-Line Encyclopedia of Integer Sequences, A014500, restricted two-covers. <https://oeis.org/A014500>. Entry and b-file retrieved 5 October 2026.
- [5] OEIS Foundation Inc., The On-Line Encyclopedia of Integer Sequences, A132219. <https://oeis.org/A132219>. The retrieved formula and b-file match the legacy series  $O$ , as discussed in this report.
- [6] NIST Digital Library of Mathematical Functions, Section 4.13, Lambert  $W$ -function, and Section 5.11, asymptotic expansions of the gamma function. The finite real Stirling remainder, not convergence of an infinite Stirling series, is used here.
- [7] ProveIt, *Transseries and inversion*, Analysis/Transseries/docs/series-and-transseries/Transseries\_And\_Inversion/transseries\_and\_inversion.tex; Part 0 (p0:def:model, p0:thm:lambert-core, p0:def:three-inverses, p0:thm:staircase) and the chapter on the Lambert template (plt:def:lw-monomial-log-datum, plt:thm:lw-template). [Added in the write.]